

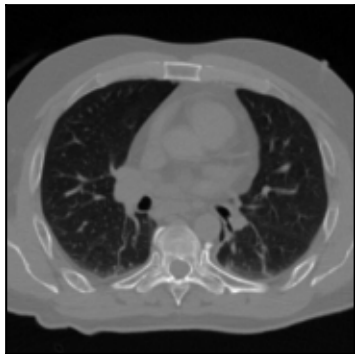
DVF QA pipeline report

DIR-Lab-case4 [frame_diff]

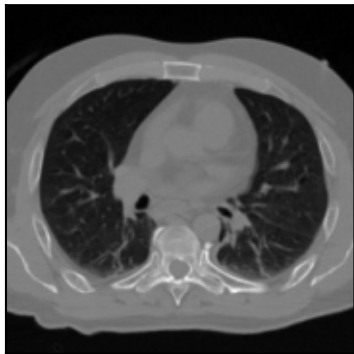
Field	Value
Case	DIR-Lab-case4 [frame_diff]
Generated	2026-05-20 08:03:09
source	Case4Pack
lateral_strategy	frame_diff
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(99, 256, 256)
spacing_xyz_mm	(1.13, 1.13, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)

4DCT phase preview

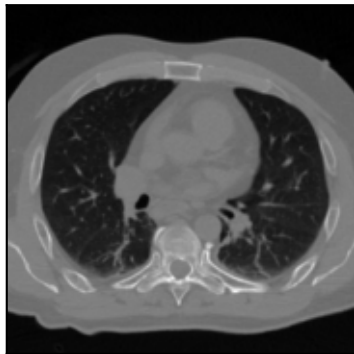
ϕ_0



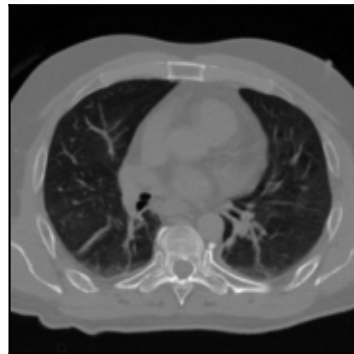
ϕ_1



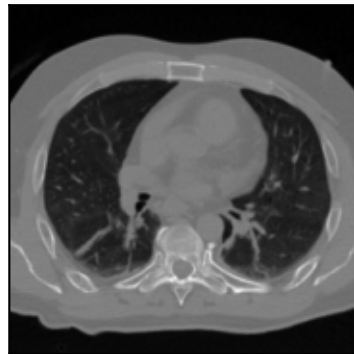
ϕ_2



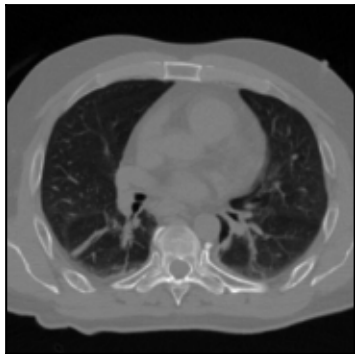
ϕ_3



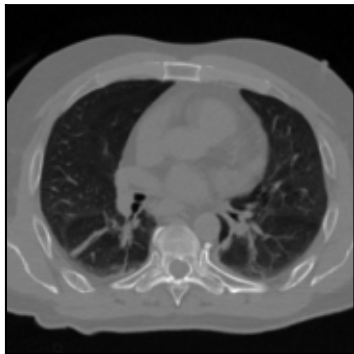
ϕ_4



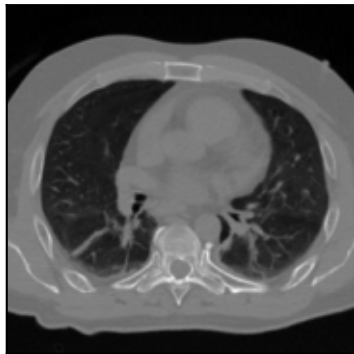
ϕ_5



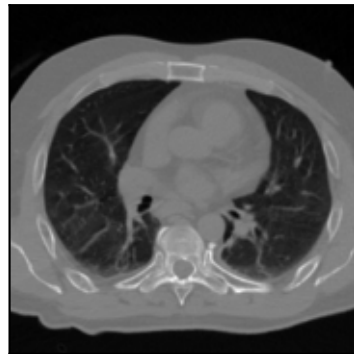
ϕ_6



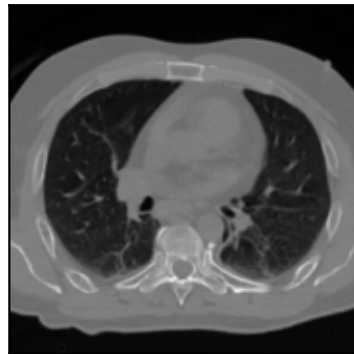
ϕ_7



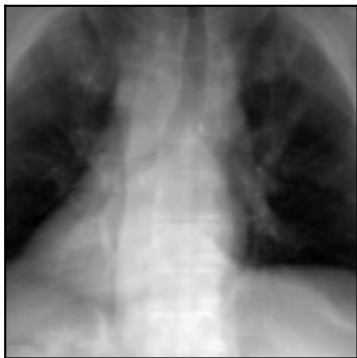
ϕ_8



ϕ_9



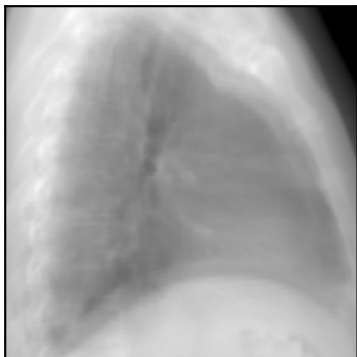
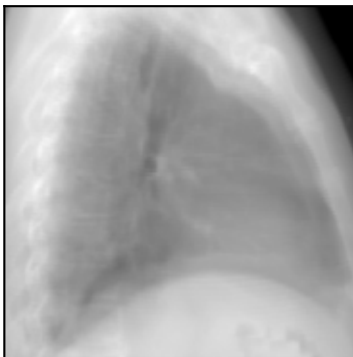
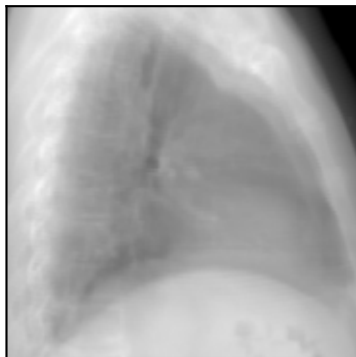
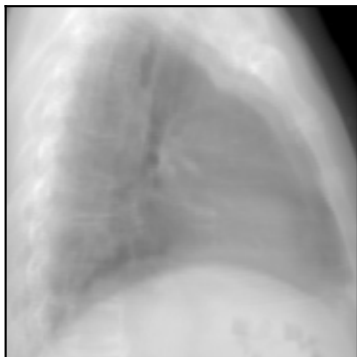
Per-phase DRRs

AP φ_0 AP $\varphi 1$ AP ϕ_2 AP $\phi 3$ AP ϕ_4 

AP $\phi 5$



AP φ6

AP $\phi 7$ AP $\phi 8$ AP ϕ_9 Lat ϕ_0 Lat ϕ_1 Lat ϕ_2 Lat ϕ_3 Lat ϕ_4 Lat ϕ_5 Lat ϕ_6 Lat ϕ_7 Lat φ_8 Lat ϕ_9 

Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



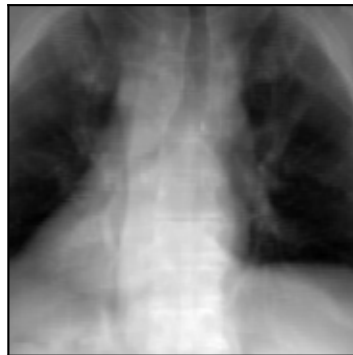
AP t=72 (4.80s)



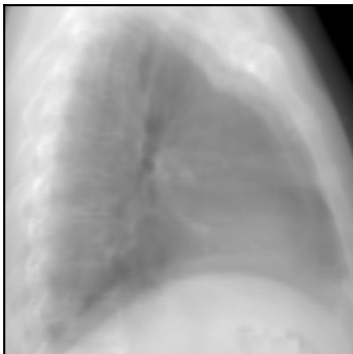
AP t=96 (6.40s)



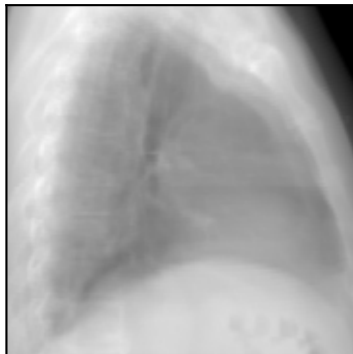
AP t=120 (8.00s)



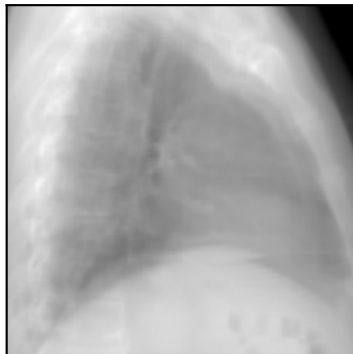
Lat t=0



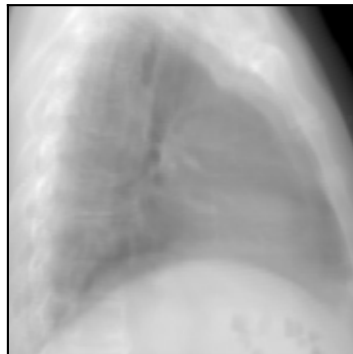
Lat t=24



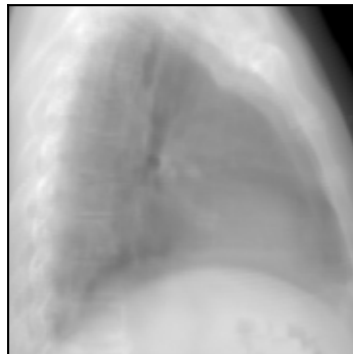
Lat t=48



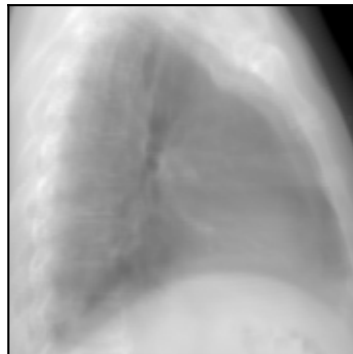
Lat t=72



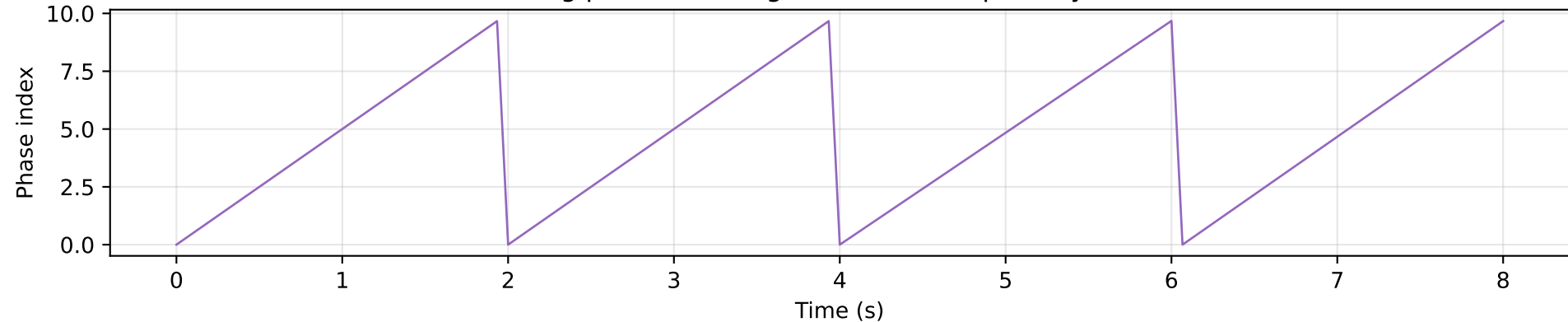
Lat t=96



Lat t=120

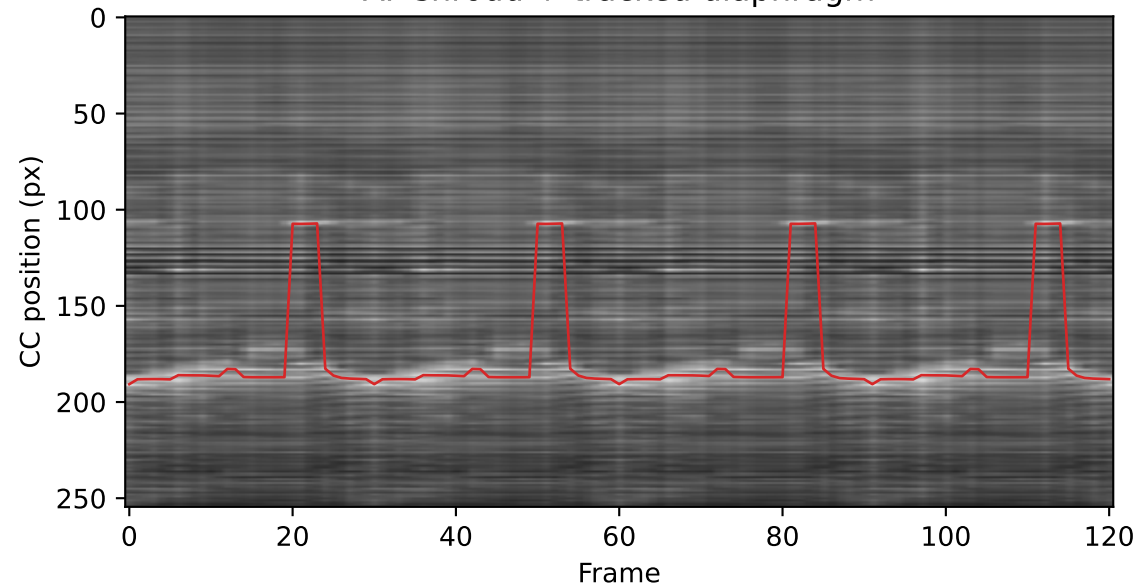


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

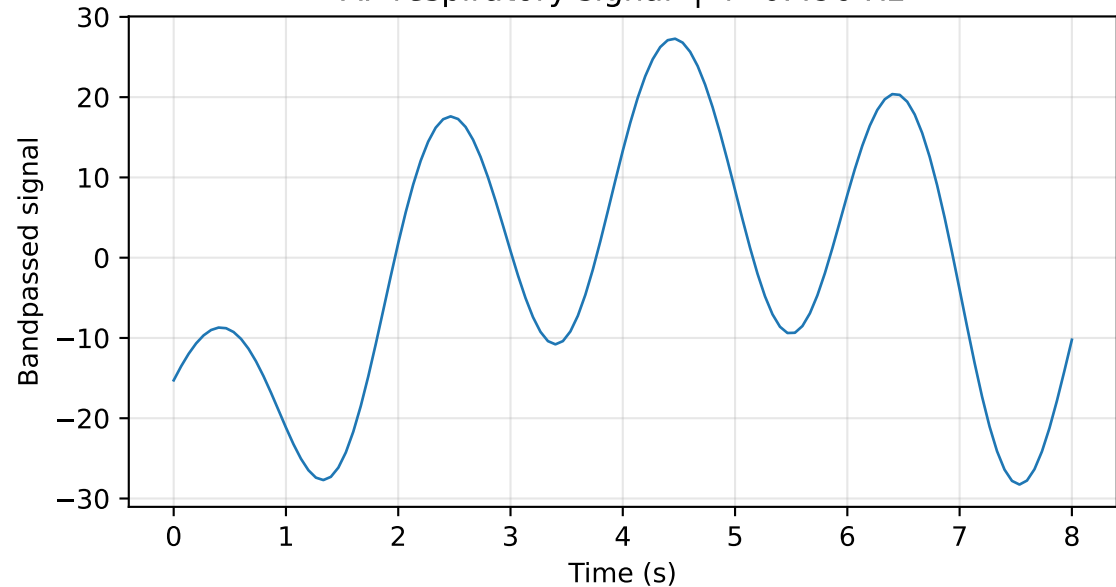


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

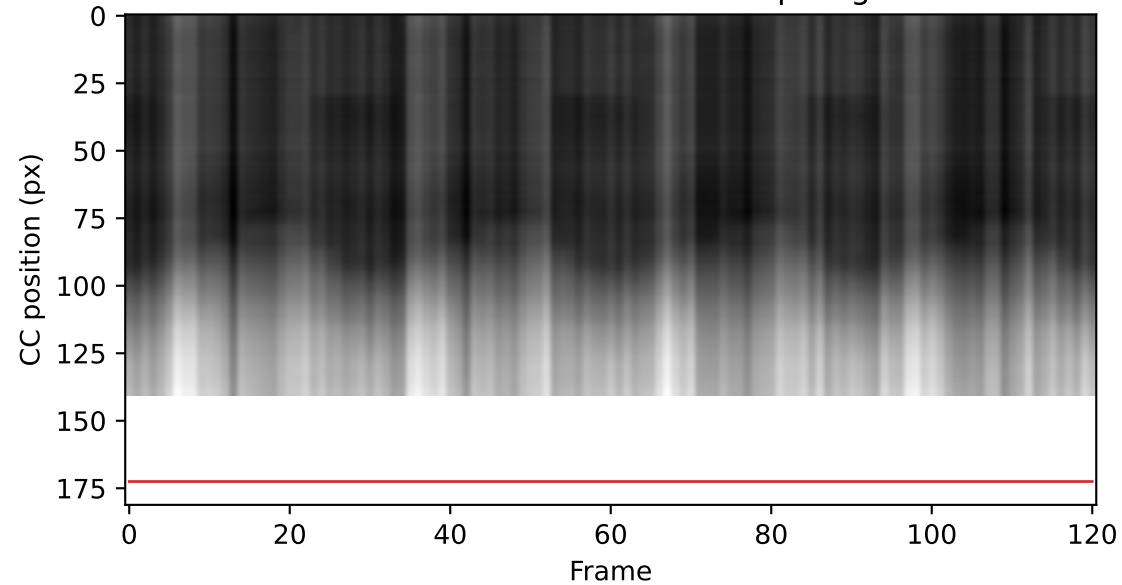


AP respiratory signal | $f=0.496$ Hz

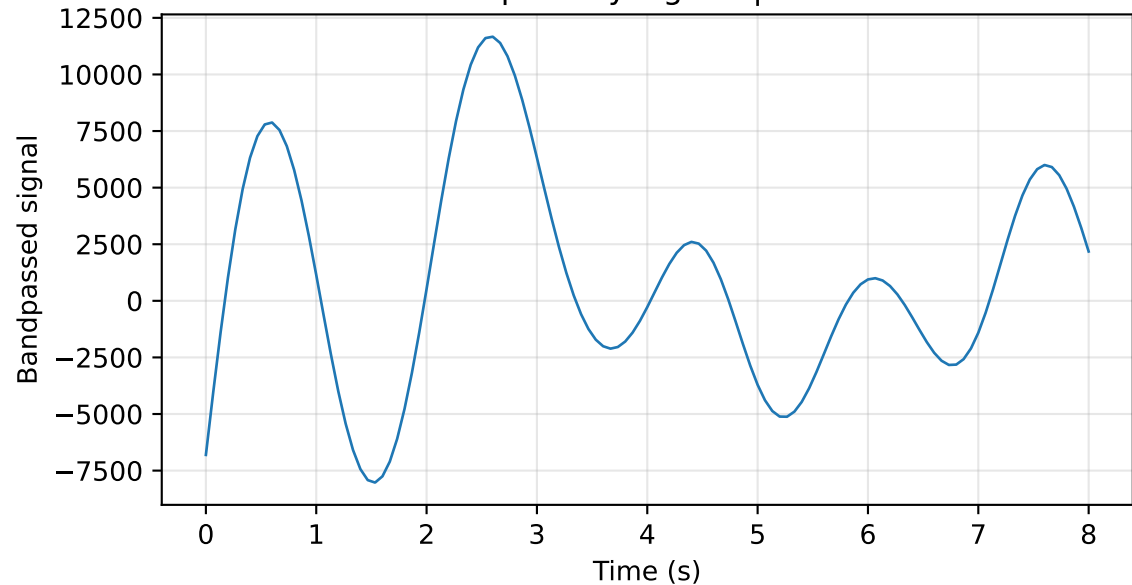


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

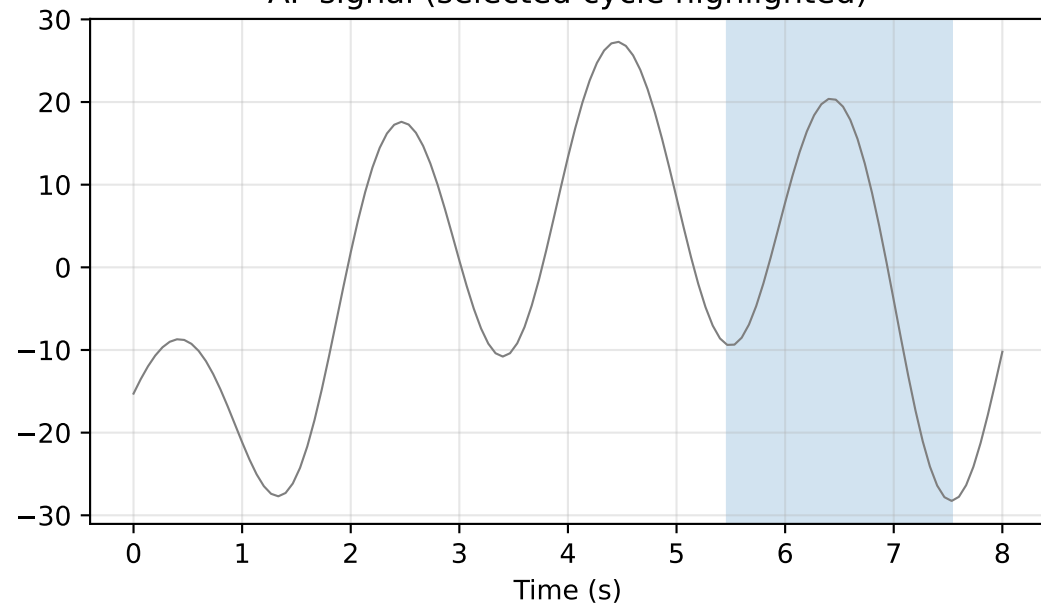


Lateral respiratory signal | $f=0.620$ Hz

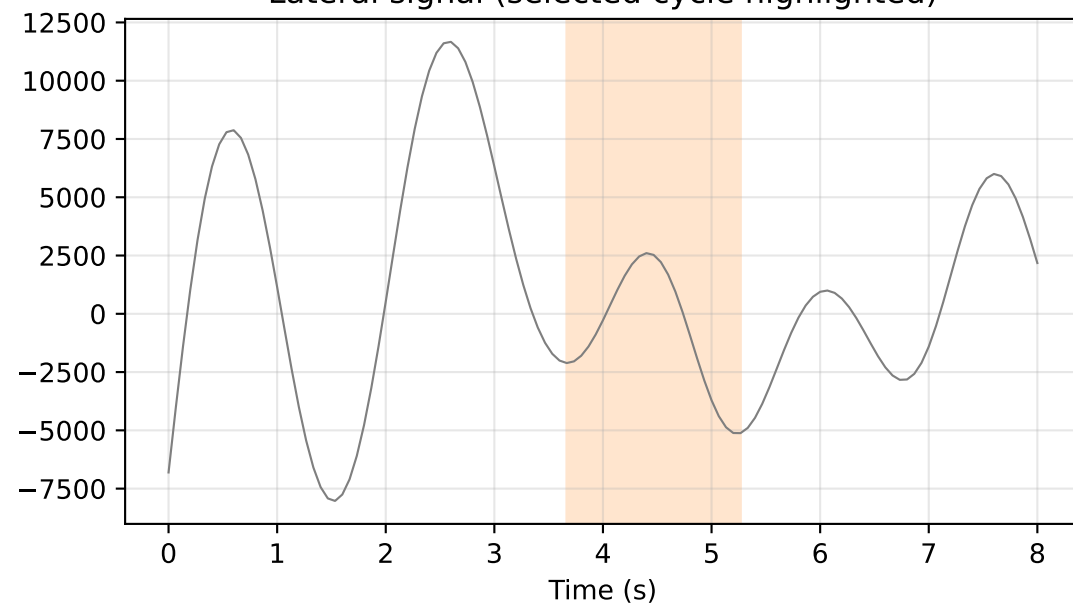


Cycle pairing

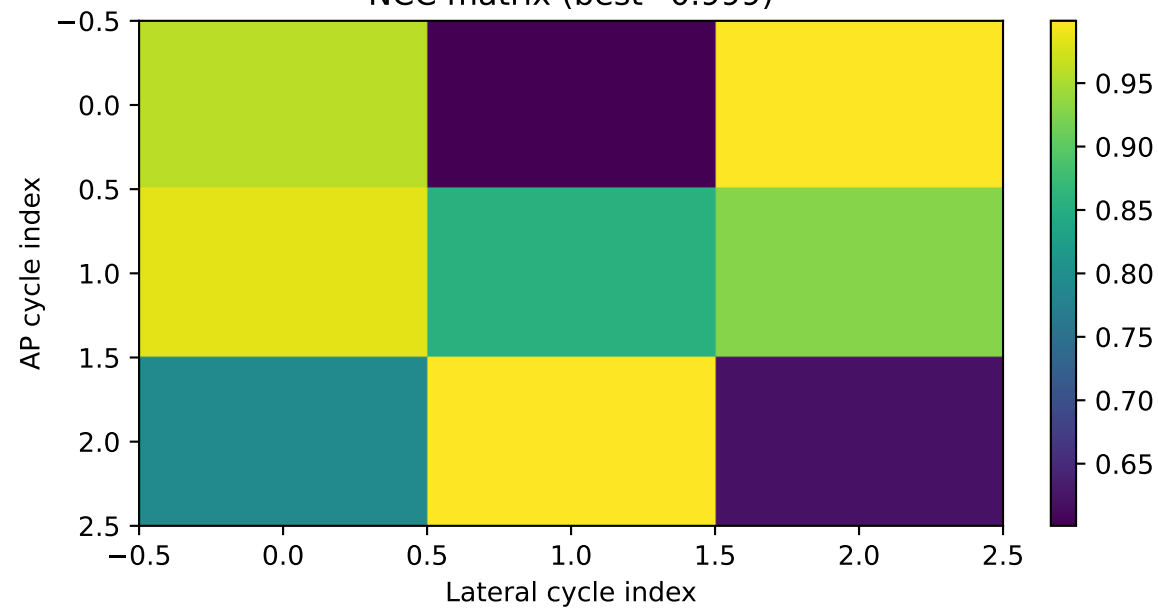
AP signal (selected cycle highlighted)



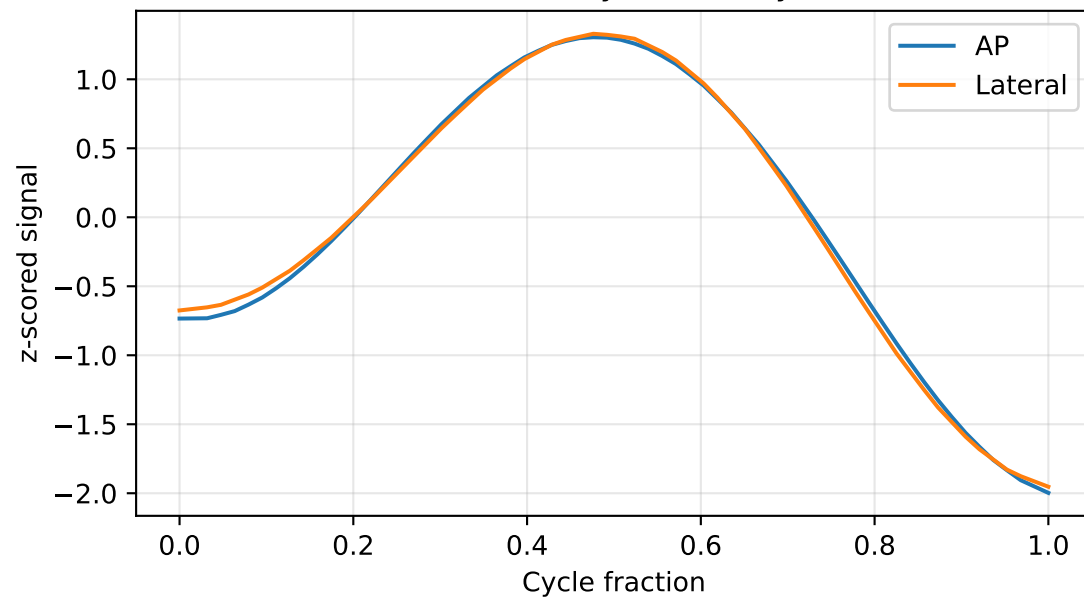
Lateral signal (selected cycle highlighted)



NCC matrix (best=0.999)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.4959
AP period (frames)	30.25
Lateral dominant frequency (Hz)	0.6198
Lateral period (frames)	24.20
Best pair NCC	0.9991
AP cycle frames	82 - 113
Lateral cycle frames	55 - 79